

# MUFBVAR

## Version

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# **Welcome to MUFBVAR's documentation!**

# Introduction

**MUFBVAR** is a module to handel, disaggregate and forecast Multiple frequency data using Bayesian VAR Models.

## Installation:

You can use pip to install the module form Gitea: .. code-block:: bash

```
foo @ bar :~$ pip install git+https://gitea.efv.admin.ch/efv_fs/MUFBVAR.git
```

# MUFBVAR

## MUFBVAR package

### Module contents

`class MUFBVAR.mufbvar_data ( data , trans , frequencies ) [source]`

Bases: `object`

Class to prepare the data that will be used in the MUFBVAR ...

Parameters :

- **data** ( *list of pandas DataFrames* ) – Data of each frequency stored in a pandas DataFrame, all stored in one list
- **trans** ( *list of numpy arrays* ) – A separate numpy array for each frequency all stored in a list. /n 0: log is taken 1: divided by 100
- **frequencies** ( *List of the frequencies of the data , in order lowest to highest* ) – “Y”, “Q”, “M”, “W”, “D”

`class MUFBVAR.multifrequency_var ( nsim , nburn_perc , nlags , thinning ) [source]`

Bases: `object`

MUFBVAR class

Parameters :

- **nsim** ( *int* ) – Number of simulations
- **nburn\_perc** ( *numeric* ) – Between 0 and 1, proportion of simulations to throw away as burn in.
- **nlags** ( *int* ) – Number of lags in the highest frequency

- **thining** ( *int* ) – To save only every nth draw

#### **aggregate** ( *frequency* , *reset\_index = True* )

Aggregates the Mean, Median and quantiles in the highest frequency to the desired frequency.

The Function ensures, that we start at the beginning of a Year or Quarter depending on the chosen frequency

##### Parameters :

- **frequency** ( *str* ) – The frequency to which the data should be aggregated to
- **reset\_index** ( *boolean* ) – Should index be changed to period Index

#### **compare\_models** ( *multifrequency\_var\_models* , *model\_names* , *agg = True* , *variables = 'all'* , *save = True* , *name = 'Comparison'* , *show = True* , *nhist = 5* )

Creates a plot with the different scenarios

##### Parameters :

- **multifrequency\_var\_models** ( *list* ) – list of multifrequency\_var\_models with forecasts
- **model\_names** ( *list* ) – list of names of the models, including the current model at position 0
- **agg** ( *boolean* ) – should the aggregated time series be shown
- **variable** ( *list of strings* ) – variables for which the plot should be generated, all if it should be generated for all
- **save** ( *boolean* ) – Whether the plots should be saved. The default is True.
- **name** ( *string* , *optional* ) – If the plots should be saved, path/name not including filetype. The default is None.
- **show** ( *boolean* ) – Whether the plots should be shown. Default is True.

- **agg** – Whether the aggregated values should be shown
- **nhist** ( *int* ) – number of historical periods that should be shown on the plot Default is 5

**fanchart** ( *variables = 'all'* , *save = True* , *name = 'Fancharts'* , *show = True* , *agg = False* , *nhist = 10* )

Creates fan plots of the desired variables.

**Parameters :**

- **variable** ( *list of strings* ) – variables for which the plot should be generated, all if it should be generated for all
- **save** ( *boolean* ) – Whether the plots should be saved. The default is True.
- **name** ( *string , optional* ) – If the plots should be saved, path/name not including filetype. The default is None.
- **show** ( *boolean* ) – Whether the plots should be shown. Default is True.
- **agg** ( *boolean* ) – Whether the aggregated values should be shown
- **nhist** ( *int* ) – number of historical periods that should be shown on the plot Default is 5

**fit** ( *mufbvar\_data* , *hyp* )

Estimates the model using the model parameter specified in the initialization.

And the data provided.

**Parameters :**

- **mufbvar\_data** ( *mufbvar\_data class object* ) – data in the form of a mufbvar\_data class object
- **hyp** ( *list of list* ) –

list containing list of the hyperparameters for each frequency step

1. overall tightness
2. scaling down the variance for the coefficients of a distant lag
3. number of observations used for obtaining the prior for the covariance matrix of error terms
4. tuning parameter for coefficients for constant
5. tuning parameter for the covariance between coefficients

**forecast ( *H* , *conditionals = None* )**

Method to generate the forecasts in the highest frequency.

**Parameters :**

- *H* ( *int* ) – Forecast horizon in highest frequency
- **conditionals** ( *pandas DataFrame or None* ) –

Conditional forecasts

column names must be the variable names

no index needed

either values or np.nan

**mean\_plot ( *variables = 'all'* , *save = True* , *name = 'Output'* , *show = True* )**

Creates cumulative mean plots of the forecasts. If the model has converged the cumulative mean should be stable after burnin.

**Parameters :**

- **variable** ( *list of strings* ) – variables for which the plot should be generated, all if it should be generated for all
- **save** ( *boolean* ) – Whether the plots should be saved. The default is True.



- **name** ( *string* , *optional* ) – If the plots should be saved, path/name not including filetype. The default is None.
- **show** ( *boolean* ) – Whether the plots should be shown. Default is True.

**save** ( *filename* = 'mufbvar\_model.pkl' )

Saves the MFBVAR Object

**Parameters :**

**filename** ( *str* ) – Path where to save the object. End must be .pkl

**scenario\_forecast** ( *H* , *conditionals* , *names* , *agg* = *True* )

Generates the mean forecasts for multiple scenarios

**Parameters :**

- **H** ( *int* ) – Forecast Horizon in highest frequency
- **conditionals** ( *list* ) – list of panda DataFrames
- **names** ( *list* ) – list of the names for the scenarios
- **agg** ( *boolean* ) –  
If true aggregates output to lowest frequency  
Default is True

**Returns :**

**out\_dict** – Dictionary containing the panda DataFrames for each szenario

**Return type :**

dict

**scenario\_plot** ( *scenario\_dict* , *variables* = 'all' , *save* = *True* , *name* = 'Scenario' , *show* = *True* , *nhist* = 5 )

Creates a plot with the different scenarios

**Parameters :**

- **scenario\_dict** ( *dict* ) – output of self.scenario\_forecast
- **variable** ( *list of strings* ) – variables for which the plot should be generated, all if it should be generated for all
- **save** ( *boolean* ) – Whether the plots should be saved. The default is True.
- **name** ( *string , optional* ) – If the plots should be saved, path/name not including filetype. The default is None.
- **show** ( *boolean* ) – Whether the plots should be shown. Default is True.
- **agg** ( *boolean* ) – Whether the aggregated values should be shown
- **nhist** ( *int* ) – number of historical periods that should be shown on the plot Default is 5

**to\_excel ( *filename* , *agg = False* )**

Writes the results to an excel

**Parameters :**

- **agg** ( *Boolean* ) – Should the aggregated series be saved
- **filename** ( *String* ) – file path.

**update\_hyperparameters ( *mufbvar\_data* , *pbounds* , *init\_points* , *n\_iter* , *nsim* , *save = False* , *name = 'hyp.txt'* )**

This method uses bayesian optimization to find the hyperparameters with the highest mdd

lambda 1: overall tightness

lambda 2: scaling down the variance for the coefficients of a distant lag

lambda 3: number of observations used for obtaining the prior for the covariance matrix of error terms, fixed to 1

lambda 4: . tuning parameter for coefficients for constant

lambda 5: tuning parameter for the covariance between coefficients

#### Parameters :

- **mufbvar\_data** ( *mufbvar\_data class object* ) – data in the form of a mufbvar\_data class object
- **pbound** ( *dict* ) –  
boundries for each hyperparameter  
  
two frequencies: lambda1\_1, lambda2\_1, lambda4\_1, lambda5\_1  
three frequencies: lambda1\_1, lambda2\_1, lambda4\_1, lambda5\_1, lambda1\_2, lambda2\_2, lambda4\_2, lambda5\_2  
four frequencies: lambda1\_1, lambda2\_1, lambda4\_1, lambda5\_1, lambda1\_2, lambda2\_2, lambda4\_2, lambda5\_2, lambda1\_3, lambda2\_3, lambda4\_3, lambda5\_3
- **init\_points** ( *int* ) – How many steps of random exploration you want to perform
- **n\_iter** ( *int* ) – How many steps of bayesian optimization you want to perform
- **nsim** ( *int* ) – number of draws in each MUFbVAR estimation
- **save** ( *boolean* ) – True if you want to save the hyperparameters as a txt
- **name** ( *str* ) – path where you want to save the hyperparameters

#### Returns :

**hyp** – list containing the optimized hyperparameters

#### Return type :

list

# Examples

Example in python:

```
import MUFBVAR
import pandas as pd
import numpy as np
import pickle

# Preparations
#-----

io_data = "hist.xlsx"

#Model Specification
H = 96          # forecast horizon
nsim = 100      # number of draws from Posterior Density
nburn = 0.5     # number of draws to discard
nlags = [6,4]   # Number of lags
thining = 1     # Thining

hyp = [[0.09, 4.3, 1, 2.7, 4.3], [0.09, 4.3, 1, 2.7, 4.3]] #
Hyperparameters see documentation for details

frequencies = ["Q","M","W"] # Frequencies

# Load the data
# -----
data = []
for freq in range(len(frequencies)):
    freq = frequencies[freq]
    data_temp = pd.read_excel(io_data, sheet_name = freq,
index_col = 0)
    data.append(data_temp)

#Transformations
trans = [np.array((1)), np.array((1,1,1)), np.array((1,1,1,1))]

# Initialize data class
data_in = MUFBVAR.mufbvar_data(data, trans, frequencies)
```

```

# Fit and Forecast
#-----

# Initialize model class
model = MUFBVAR.multifrequency_var(nsim, nburn, nlags ,thining)

# Estimate the model
model.fit(data_in, hyp = hyp)

# Conditional forecasts

conditionals = pd.DataFrame({'w_1' : [0.018, 0.025, np.nan, np.nan,
0.0228, 0.05],
                             'm_2' : [ np.nan, 0.002, 0.01 , 0.01, np.nan,
np.nan]})

# Create forecasts in highest frequency
model.forecast(H, conditionals)

# Aggregate
model.aggregate(frequency = "Q")

# Save results
#-----
#model.to_excel('out_test.xlsx', agg = True)
#model.save("model_2002_Q4")

# Plots
#-----

model.mean_plot(variables = "all", save = False, show = True)

model.fanchart(variables = "all", save = False, show = True, agg =
True, nhist = 10)

# Optimizing Hyperparameters
#-----

# Define boundaries for each hyperparameter, see documentation for
details
pbounds = {'lambda1_1': (0.001, 20), 'lambda2_1': (0.01, 10),
'lambda4_1': (0.01, 10), 'lambda5_1': (0.01, 10), 'lambda1_2':
(0.001, 20), 'lambda2_2': (0.01, 10), 'lambda4_2': (0.01, 10),
'lambda5_2': (0.01, 10)}
init_points = 3 # number of random points
n_iter = 8 # number of baysian optimization steps
nsim = 100 # number of simulations

hyp = model.update_hyperparameters(data_in, pbounds, init_points,
n_iter, nsim, save = False, name = "hyp.txt")

```

```

# Scenario Analysis
#-----

# We can compare different scenarios

conditionals = [pd.DataFrame({'w_1' : [0.018, 0.025, np.nan, np.nan,
0.0228, 0.05],
                             'm_2' : [ 0.3, 0.002, 0.01 , 0.01, np.nan,
np.nan]}),
                pd.DataFrame({'w_1' : [-0.02, -0.25, np.nan, np.nan,
-0.228, 0.1],
                             'm_2' : [ -0.2, -0.012, 0 , 0.1, np.nan,
np.nan]}),
                None
                ]

names = ["good", "bad", "base"]

out_scenarios = model.scenario_forecast(H, conditionals, names, agg =
True)

# Scenario Plot
model.scenario_plot(scenario_dict = out_scenarios, variables = "all",
save = False, name = "Scenario", show = True, nhist = 10)

# Compare with model from last quarter
#-----

#load previous quarter model:
file = open("model_2001_Q3.pkl",'rb')
model_2001_Q3 = pickle.load(file)

model_names = ["2001-Q4", "2001-Q3"]
multifrequency_var_models = [model_2001_Q3]

model.compare_models(multifrequency_var_models, model_names, agg =
True, variables = "all", save = False, name = "Comparison", show =
True, nhist = 5)

model.compare_models(multifrequency_var_models, model_names, agg =
False, variables = ["q_1"], save = False, name = "Comparison", show =
True, nhist = 20)

```

## Example in R:

The module can also be used in r, using the reticulate package. The easiest way is to source a python script with reticulate

```
library(reticulate)
setwd(dirname(rstudioapi::getActiveDocumentContext())$path))

source_python('python_example.py')
```

It is also possible to load the package into R and use the functions inside of an R script.

```
library(reticulate)

setwd(dirname(rstudioapi::getActiveDocumentContext())$path))

mufbvar <- import("MUFBVAR")
pd <- impoty("pandas")
np <- import("numpy")
pickle <- import("pickle")

io_data <- "hist.xlsx"

# Preparations
#-----

H <- 96L           # forecast horizon
nsim <- 20000L     # number of draws from Posterior Density
nburn <- 0.5       # number of draws to discard
nlags <- list(6L,4L)
thining = 1

hyp <- list(list(0.09, 4.3, 1, 2.7, 4.3), list(0.09, 4.3, 1, 2.7,
4.3))

frequencies <- list("Q", "M", "W")

data <- list()
for (freq in 1:length(frequencies)) {
  freq <- frequencies[[freq]]
  data_temp <- pd$read_excel(io_data, sheet_name = freq, index_col
= 0)
  data <- append(data, list(data_temp))
}

#Transformations
trans <- list(np$array((1)), np$array((1,1,1)), np$array((1,1,1,1)))

#Initialize data class
data_in <- mufbvar$mufbvar_data(data, trans, frequencies)

# Fit and Forecast
```

```

#-----

# Initialize model class
model <- MUFbVAR$multifrequency_var(nsim, nburn, nlags ,thining)

# Estimate the model
model$fit(data_in, hyp = hyp)

# Conditional forecasts

conditionals <- pd$DataFrame({'w_1' : [0.018, 0.025, np.nan, np.nan,
0.0228, 0.05],
                             'm_2' : [ np.nan, 0.002, 0.01 , 0.01,
np.nan, np.nan]})

# Create forecasts in highest frequency
model$forecast(H, conditionals)

# Aggregate
model$aggregate(frequency = "Q")

# Save results
#-----
#model$to_excel('out_test.xlsx', agg = True)
#model$save("model_2002_Q4")

# Plots
#-----

model$mean_plot(variables = "all", save = False, show = True)

model$fanchart(variables = "all", save = False, show = True, agg =
True, nhist = 10)

# Optimizing Hyperparameters
#-----

# Define boundaries for each hyperparameter, see documentation for
details
pbounds <- {'lambda1_1': (0.001, 20), 'lambda2_1': (0.01, 10),
'lambda4_1': (0.01, 10), 'lambda5_1': (0.01, 10), 'lambda1_2':
(0.001, 20), 'lambda2_2': (0.01, 10), 'lambda4_2': (0.01, 10),
'lambda5_2': (0.01, 10)}
init_points <- 3L # number of random points
n_iter <- 8L # number of bayesian optimization steps
nsim <- 100L # number of simulations

hyp <- model$update_hyperparameters(data_in, pbounds, init_points,
n_iter, nsim, save = False, name = "hyp.txt")

# Scenario Analysis
#-----

```



```

# We can compare different scenarios

conditionals <- [pd$DataFrame({'w_1' : [0.018, 0.025, np.nan,
np.nan, 0.0228, 0.05],
                             'm_2' : [ 0.3, 0.002, 0.01 , 0.01, np.nan,
np.nan]}),
                pd$DataFrame({'w_1' : [-0.02, -0.25, np.nan, np.nan,
-0.228, 0.1],
                             'm_2' : [ -0.2, -0.012, 0 , 0.1, np.nan,
np.nan]}),
                None
                ]

names <- list("good", "bad", "base")

out_scenarios = model$scenario_forecast(H, conditionals, names, agg
= True)

# Scenario Plot
model$scenario_plot(scenario_dict = out_scenarios, variables =
"all", save = False, name = "Scenario", show = True, nhist = 10L)

# Compare with model from last quarter
#-----

#load previous quarter model:
file <- open("model_2001_Q3.pkl",'rb')
model_2001_Q3 <- pickle.load(file)

model_names <- list("2001-Q4", "2001-Q3")
multifrequency_var_models <- list(model_2001_Q3)

model$compare_models(multifrequency_var_models, model_names, agg =
True, variables = "all", save = False, name = "Comparison", show =
True, nhist = 5L)

model$compare_models(multifrequency_var_models, model_names, agg =
False, variables = ["q_1"], save = False, name = "Comparison", show
= True, nhist = 20L)

```

# Indices and tables

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